

Qubble: Numerical Implementation Notes

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Abstract

These notes describe the numerical implementation of **Qubble**, the code used to construct fixed-charge tunnelling solutions in the companion paper. The problem is the semiclassical decay of a homogeneous medium carrying a conserved global $U(1)$ charge. At fixed charge, the Euclidean path integral involves twisted boundary conditions and the saddle is naturally formulated in terms of two independent Euclidean fields. We present the conventions used in the code, the rotated variables that remove the homogeneous Euclidean exponential, the discretized boundary-value problem, the Newton strategy used to solve it, and the fixed-charge outer loop that determines the residual twist. We also describe the construction of the initial ansatz families and the continuation of the turning slice to real time, including the energy and charge diagnostics used to validate the solution. The aim is to provide a self-contained technical reference for reproducing and extending the numerical results.

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1 Overview and physical problem

These notes document the numerical implementation used in **Qubble**. The code solves the semiclassical bubble-nucleation problem at fixed conserved global charge Q , for the case of a complex scalar field with a global $U(1)$ symmetry. The goal is to construct the Euclidean saddle that describes the decay of a homogeneous charged metastable state, compute its fixed- Q suppression exponent, and use the turning slice of the Euclidean solution as initial data for the subsequent Minkowski evolution.

The starting point is a relativistic complex scalar field,

$$\Phi(t, \mathbf{x}) = \frac{\rho(t, \mathbf{x})}{\sqrt{2}} e^{i\theta(t, \mathbf{x})}, \quad (1.1)$$

with conserved $U(1)$ charge

$$Q = \int d^3x j^0, \quad j^0 = i(\Phi^* \dot{\Phi} - \dot{\Phi}^* \Phi) = \rho^2 \dot{\theta}. \quad (1.2)$$

The homogeneous charged state around which the decay problem is formulated is a rigidly rotating configuration,

$$\Phi_i(t) = \frac{\rho_i}{\sqrt{2}} e^{i\omega t}, \quad q_i = \omega \rho_i^2. \quad (1.3)$$

The modulus ρ_i is fixed by the homogeneous equation of motion,

$$\left. \frac{dU}{d\rho} \right|_{\rho=\rho_i} = \omega^2 \rho_i, \quad (1.4)$$

where $U(\rho)$ is the scalar potential written as a function of the physical modulus ρ .

At zero charge, false-vacuum decay is described by the ordinary real Coleman bounce, which is $O(4)$ -symmetric at zero temperature. At fixed nonzero charge, the problem is structurally different. The tunnelling event must take place inside a definite charge sector, so the Euclidean path integral must be projected onto fixed Q . Formally,

$$Z_Q(\beta) = \text{Tr}_Q e^{-\beta \hat{H}} = \int \frac{d\theta}{2\pi} e^{-i\theta Q} \text{Tr} \left(e^{-\beta \hat{H}} e^{i\theta \hat{Q}} \right). \quad (1.5)$$

The operator $e^{i\theta \hat{Q}}$ acts as a global $U(1)$ rotation. Thus, in the Euclidean path integral, the fields are not periodic in Euclidean time, but obey twisted boundary conditions. At the saddle, the relevant twist is imaginary, $\theta = -i\eta_{\text{cl}}$, so that the Euclidean boundary conditions become real exponential conditions,

$$\phi(\tau + \beta, r) = e^{\eta_{\text{cl}}} \phi(\tau, r), \quad \bar{\phi}(\tau + \beta, r) = e^{-\eta_{\text{cl}}} \bar{\phi}(\tau, r). \quad (1.6)$$

Here ϕ and $\bar{\phi}$ are independent Euclidean fields. They should not be identified with one another away from the turning slice.

This independence is not a numerical trick. It is the real Euclidean section of the complex saddle selected by the fixed-charge projection. After Wick rotation, the homogeneous charged background (1.3) becomes

$$\phi_i(\tau) = \frac{\rho_i}{\sqrt{2}} e^{+\omega \tau}, \quad \bar{\phi}_i(\tau) = \frac{\rho_i}{\sqrt{2}} e^{-\omega \tau}. \quad (1.7)$$

Therefore, the Euclidean “phase” is hyperbolic rather than oscillatory. A single real Euclidean field cannot describe this configuration while keeping the charge real. The correct saddle-point problem is instead formulated in terms of two independent fields ϕ and $\bar{\phi}$, which are matched back to a physical complex field only on the turning slice.

The Euclidean action used in the numerical problem is

$$S_E[\phi, \bar{\phi}] = \int_{-\beta/2}^{+\beta/2} d\tau \int d^3x \left[\partial_\tau \phi \partial_\tau \bar{\phi} + \nabla \phi \cdot \nabla \bar{\phi} + U(\rho) \right], \quad \rho = \sqrt{2\phi\bar{\phi}}. \quad (1.8)$$

Varying independently with respect to ϕ and $\bar{\phi}$ gives

$$\partial_\tau^2 \phi + \nabla^2 \phi - \frac{1}{\rho} \frac{dU}{d\rho} \phi = 0, \quad (1.9)$$

$$\partial_\tau^2 \bar{\phi} + \nabla^2 \bar{\phi} - \frac{1}{\rho} \frac{dU}{d\rho} \bar{\phi} = 0. \quad (1.10)$$

Assuming spherical symmetry in the spatial directions, the fields depend only on (τ, r) , and the equations become

$$\partial_\tau^2 \phi + \partial_r^2 \phi + \frac{2}{r} \partial_r \phi - \frac{1}{\rho} \frac{dU}{d\rho} \phi = 0, \quad (1.11)$$

$$\partial_\tau^2 \bar{\phi} + \partial_r^2 \bar{\phi} + \frac{2}{r} \partial_r \bar{\phi} - \frac{1}{\rho} \frac{dU}{d\rho} \bar{\phi} = 0. \quad (1.12)$$

These are the continuum equations whose discretized version is solved by the code.

The boundary conditions are part of the definition of the fixed- Q bounce. Regularity at the origin requires

$$\partial_r \phi(\tau, 0) = 0, \quad \partial_r \bar{\phi}(\tau, 0) = 0. \quad (1.13)$$

At large radius, the solution must approach the homogeneous charged background,

$$\phi(\tau, r) \xrightarrow{r \rightarrow \infty} \phi_i(\tau), \quad \bar{\phi}(\tau, r) \xrightarrow{r \rightarrow \infty} \bar{\phi}_i(\tau). \quad (1.14)$$

In Euclidean time, the full solution obeys the twisted closure (1.6). Finally, the tunnelling solution has a turning slice, chosen to be at $\tau = 0$, where the Euclidean fields are matched to physical Minkowski initial data. The turning conditions are

$$\phi(0, r) = \bar{\phi}(0, r), \quad \partial_\tau \phi(0, r) = -\partial_\tau \bar{\phi}(0, r). \quad (1.15)$$

These conditions guarantee that after analytic continuation, one obtains a single physical complex field satisfying the usual conjugation relation.

The solution has the reflection-and-exchange symmetry

$$\phi(\tau, r) = \bar{\phi}(-\tau, r), \quad \bar{\phi}(\tau, r) = \phi(-\tau, r). \quad (1.16)$$

For this reason the code only solves the problem on the half-domain

$$r \in (0, L_r), \quad \tau \in \left(-\frac{\beta}{2}, 0\right), \quad (1.17)$$

and reconstructs the $\tau > 0$ branch by (1.16).

For numerical stability, the homogeneous exponential factors in (1.7) are removed by working in a rotated Euclidean frame,

$$\phi(\tau, r) = e^{+\omega\tau} \phi_{\text{rot}}(\tau, r), \quad \bar{\phi}(\tau, r) = e^{-\omega\tau} \bar{\phi}_{\text{rot}}(\tau, r). \quad (1.18)$$

In this frame, the homogeneous background is constant. The total twist η_{cl} is correspondingly split into the trivial homogeneous part $\omega\beta$ and a residual twist,

$$\eta_0 = \eta_{\text{cl}} - \omega\beta. \quad (1.19)$$

The rotated fields obey

$$\phi_{\text{rot}}(\tau + \beta, r) = e^{\eta_0} \phi_{\text{rot}}(\tau, r), \quad \bar{\phi}_{\text{rot}}(\tau + \beta, r) = e^{-\eta_0} \bar{\phi}_{\text{rot}}(\tau, r). \quad (1.20)$$

The code then introduces regularized unknowns

$$y(\tau, r) = r [\phi_{\text{rot}}(\tau, r) - \varphi_i], \quad \bar{y}(\tau, r) = r [\bar{\phi}_{\text{rot}}(\tau, r) - \varphi_i], \quad \varphi_i = \frac{\rho_i}{\sqrt{2}}. \quad (1.21)$$

The factor of r removes the explicit first-derivative term in the spherical Laplacian and maps the homogeneous solution to

$$y = \bar{y} = 0. \quad (1.22)$$

In these variables, the continuum residual equations take the compact form

$$0 = \partial_\tau^2 y + \partial_r^2 y + 2\omega \partial_\tau y + A(u)y_{\text{tot}}, \quad (1.23)$$

$$0 = \partial_\tau^2 \bar{y} + \partial_r^2 \bar{y} - 2\omega \partial_\tau \bar{y} + A(u)\bar{y}_{\text{tot}}, \quad (1.24)$$

where

$$y_{\text{tot}} = y + r\varphi_i, \quad \bar{y}_{\text{tot}} = \bar{y} + r\varphi_i, \quad u = \frac{y_{\text{tot}}\bar{y}_{\text{tot}}}{r^2}, \quad (1.25)$$

and

$$A(u) = \omega^2 - \frac{1}{\rho(u)} \frac{dU}{d\rho} \Big|_{\rho=\rho(u)}, \quad \rho(u) = \sqrt{2u}. \quad (1.26)$$

Equations (1.23)–(1.24) are the equations actually discretized in the Euclidean solver.

The fixed-charge condition is imposed after solving the boundary-value problem for a trial value of η_0 . The Euclidean charge density is

$$q_E(\tau, r) = \bar{\phi} \partial_\tau \phi - \phi \partial_\tau \bar{\phi}, \quad (1.27)$$

and the total charge is

$$Q(\tau) = 4\pi \int_0^{L_r} dr r^2 q_E(\tau, r). \quad (1.28)$$

For an exact solution this charge is independent of τ . Numerically, it is evaluated at the turning slice using the same ghost rules that implement (1.15). The residual twist is then tuned until

$$Q_{\text{bounce}}(\eta_0) = Q_{\text{target}}. \quad (1.29)$$

The complete numerical strategy is therefore:

1. Choose the model potential $U(\rho)$, the box size L_r , the Euclidean time extent β , and the target charge Q_{target} .
2. Determine the homogeneous charged reference state by solving (1.4). This fixes ρ_i , $\varphi_i = \rho_i/\sqrt{2}$, and the homogeneous charge in the finite box.
3. Construct one-dimensional reduced bounce profiles. These are not the final fixed- Q solution, but they provide controlled initial seeds for the full two-dimensional Newton solve.
4. Build two-dimensional seeds on the (r, τ) half-domain. The most important seed is the $O(4)$ -connected one, obtained by embedding the one-dimensional $d = 4$ profile through $s = \sqrt{r^2 + \tau^2}$. Additional $O(3)$ -static and $O(1)$ -like seeds are used to map nearby branches and identify nonphysical solutions.
5. For a trial value of the residual twist η_0 , solve the discretized nonlinear system corresponding to (1.23)–(1.24). This is done with Newton iteration, using an analytic sparse Jacobian and a line-search damping step.
6. Compute the charge of the resulting saddle using (1.28). Update η_0 and repeat the Newton solve until (1.29) is satisfied.
7. Once the fixed- Q saddle has been found, compute the suppression exponent relative to the homogeneous charged reference configuration,

$$F_{Q,\beta} = S_E[\phi_b, \bar{\phi}_b] - S_E[\phi_i, \bar{\phi}_i] + \eta_0 Q. \quad (1.30)$$

8. Check energy and charge conservation. In particular, the conserved Euclidean energy must agree between the asymptotic homogeneous state and the turning slice. This verifies that the nucleated bubble has the same total energy as the initial homogeneous charged state.

9. Use the turning slice as initial data for Minkowski evolution. The modulus is momentarily at a turning point, but the phase still carries the conserved charge. The subsequent real-time evolution determines the expanding charged bubble, the charge rearrangement, and the late-time wall velocity.

The important point is that this is a saddle-point boundary-value problem, not an energy minimization. The Newton solver can converge to several branches depending on the seed. The physical branch relevant for zero-temperature tunnelling is the one continuously connected to the neutral Coleman $O(4)$ bounce as $Q \rightarrow 0$, satisfies the fixed-charge condition, obeys the turning slice energy check, and gives a consistent, real-time expanding bubble after analytic continuation.

The purpose of the remaining sections is to spell out the implementation of this strategy in detail:

- the field conventions used in the code;
- the rotated variables that remove the homogeneous Euclidean exponential;
- the finite-difference residual equations;
- the half-box ghost relations implementing the turning slice and fixed-charge twist;
- the Newton solve and fixed-charge outer loop;
- the ansatz families used to initialize the nonlinear solve;
- the Minkowski continuation and the energy/charge diagnostics.

2 Numerical implementation

We now describe the numerical setup used to solve the Euclidean fixed-charge boundary-value problem. Since several closely related field variables appear in the analytic formulation and in the code, we first state our conventions. The independent Euclidean fields are denoted by

$$\phi(\tau, r), \quad \bar{\phi}(\tau, r), \quad (2.1)$$

and should not be identified with one another away from the turning slice. Their product defines the solver-normalized radial variable

$$\varphi(\tau, r) \equiv \sqrt{\phi(\tau, r)\bar{\phi}(\tau, r)}. \quad (2.2)$$

The physical modulus used in the main text is instead

$$\rho(\tau, r) = \sqrt{2}\varphi(\tau, r) = \sqrt{2\phi(\tau, r)\bar{\phi}(\tau, r)}. \quad (2.3)$$

Thus a homogeneous configuration with physical modulus ρ_i corresponds in the code to

$$\varphi_i = \frac{\rho_i}{\sqrt{2}}. \quad (2.4)$$

In the numerical implementation, we work directly with the fields $\phi, \bar{\phi}$ and with the solver variable φ . The physical modulus ρ is reconstructed only when needed for comparison with the notation of the main text or for plotting.

The potential is written as $U(\rho)$, with ρ the physical modulus. Therefore, whenever the code evaluates forces in terms of $\phi, \bar{\phi}$, the relation (2.3) must be used. The homogeneous charged background is

$$\phi_i(\tau) = \varphi_i e^{+\omega\tau}, \quad \bar{\phi}_i(\tau) = \varphi_i e^{-\omega\tau}, \quad (2.5)$$

so that the Euclidean charge density is $q_i = 2\omega\varphi_i^2 = \omega\rho_i^2$. The homogeneous stationarity condition is

$$\left. \frac{dU}{d\rho} \right|_{\rho=\rho_i} = \omega^2 \rho_i = \omega^2 \sqrt{2} \varphi_i. \quad (2.6)$$

Our Newton-based treatment of the fixed- Q saddle follows the same general strategy adopted in numerical studies of Q-ball tunnelling, in particular Ref. [1], but the implementation is adapted to the present two-dimensional (r, τ) problem with spherical symmetry in space.

2.1 Domain, grid, and rotated variables

We solve the Euclidean equations on the half-domain

$$r \in (0, L_r), \quad \tau \in \left(-\frac{\beta}{2}, 0\right), \quad (2.7)$$

and reconstruct the $\tau > 0$ branch by the reflection-and-exchange symmetry at the turning slice. The grid is uniform and staggered,

$$r_j = (j+1)\Delta r, \quad \tau_i = -\left(i + \frac{1}{2}\right)\Delta\tau, \quad j = 0, \dots, N_r - 1, \quad i = 0, \dots, N_\tau - 1, \quad (2.8)$$

with

$$\Delta r = \frac{L_r}{N_r}, \quad \Delta\tau = \frac{\beta}{2N_\tau}. \quad (2.9)$$

The half-step placement is useful because both $\tau = 0$ and $\tau = -\beta/2$ are implemented through ghost relations, while centred finite differences can be used in the interior.

The homogeneous Euclidean background (2.5) contains exponentially growing and decaying factors. To remove this stiffness, the code works in the rotated frame

$$\phi(\tau, r) = e^{+\omega\tau} \phi_{\text{rot}}(\tau, r), \quad \bar{\phi}(\tau, r) = e^{-\omega\tau} \bar{\phi}_{\text{rot}}(\tau, r), \quad (2.10)$$

or equivalently

$$\phi_{\text{rot}}(\tau, r) = e^{-\omega\tau} \phi(\tau, r), \quad \bar{\phi}_{\text{rot}}(\tau, r) = e^{+\omega\tau} \bar{\phi}(\tau, r). \quad (2.11)$$

In these variables, the homogeneous background is simply

$$\phi_{\text{rot},i} = \bar{\phi}_{\text{rot},i} = \varphi_i. \quad (2.12)$$

The solver unknowns are regularized fluctuations around this rotated homogeneous background,

$$y(\tau, r) = r [\phi_{\text{rot}}(\tau, r) - \varphi_i], \quad \bar{y}(\tau, r) = r [\bar{\phi}_{\text{rot}}(\tau, r) - \varphi_i]. \quad (2.13)$$

Thus

$$\phi_{\text{rot}}(\tau, r) = \varphi_i + \frac{y(\tau, r)}{r}, \quad \bar{\phi}_{\text{rot}}(\tau, r) = \varphi_i + \frac{\bar{y}(\tau, r)}{r}. \quad (2.14)$$

The homogeneous solution is mapped to

$$y = \bar{y} = 0, \quad (2.15)$$

and the factor of r in (2.13) removes the explicit first-derivative term in the spherical Laplacian.

It is useful to define the total regularized fields

$$y_{\text{tot}}(\tau, r) = y(\tau, r) + r\varphi_i, \quad \bar{y}_{\text{tot}}(\tau, r) = \bar{y}(\tau, r) + r\varphi_i, \quad (2.16)$$

so that

$$\phi_{\text{rot}} = \frac{y_{\text{tot}}}{r}, \quad \bar{\phi}_{\text{rot}} = \frac{\bar{y}_{\text{tot}}}{r}, \quad u \equiv \phi_{\text{rot}} \bar{\phi}_{\text{rot}} = \frac{y_{\text{tot}} \bar{y}_{\text{tot}}}{r^2}. \quad (2.17)$$

All nonlinear terms and all twisted boundary conditions are imposed on the total fields $y_{\text{tot}}, \bar{y}_{\text{tot}}$, not only on the fluctuations $y, \bar{y}$. This is essential: twisting only the fluctuations while leaving the homogeneous piece $r\varphi_i$ untouched would not implement the fixed-charge closure for the full fields.

The code keeps y and $\bar{y}$ independent, exactly as the continuum Euclidean formulation treats ϕ and $\bar{\phi}$ as independent fields. When complex saddles are allowed, the Newton vector is the real split

$$\mathbf{x} = (\text{Re } y, \text{Im } y, \text{Re } \bar{y}, \text{Im } \bar{y}). \quad (2.18)$$

The nonlinear system is therefore written as a real system with a real, sparse Jacobian.

2.2 Bulk residual equations

The Euclidean equations for $\phi, \bar{\phi}$, after imposing spherical symmetry and transforming to the rotated variables, can be written compactly in terms of $y, \bar{y}$. Define

$$\rho(u) = \sqrt{2u}, \quad W(u) \equiv \frac{1}{2\rho(u)} \left. \frac{dU}{d\rho} \right|_{\rho=\rho(u)}, \quad (2.19)$$

and

$$A(u) \equiv \omega^2 - 2W(u). \quad (2.20)$$

The homogeneous condition (2.6) is equivalently

$$A(\varphi_i^2) = 0. \quad (2.21)$$

In terms of these quantities, the continuum residual equations implemented by the code are

$$0 = \partial_\tau^2 y + \partial_r^2 y + 2\omega \partial_\tau y + A(u) y_{\text{tot}}, \quad (2.22)$$

and

$$0 = \partial_\tau^2 \bar{y} + \partial_r^2 \bar{y} - 2\omega \partial_\tau \bar{y} + A(u) \bar{y}_{\text{tot}}. \quad (2.23)$$

These equations are evaluated pointwise on the grid using centred second-order finite differences in the interior. The nonlinear dependence enters only through

$$u = \frac{y_{\text{tot}} \bar{y}_{\text{tot}}}{r^2}. \quad (2.24)$$

2.3 Half-box boundary conditions

The fixed-charge projection imposes quasi-periodic boundary conditions on the full Euclidean interval,

$$\phi(\tau + \beta, r) = e^{\eta_{\text{cl}}} \phi(\tau, r), \quad \bar{\phi}(\tau + \beta, r) = e^{-\eta_{\text{cl}}} \bar{\phi}(\tau, r). \quad (2.25)$$

In the rotated frame (2.10), the homogeneous contribution $\omega\beta$ is removed. Therefore, the twist that enters the numerical problem is the residual twist

$$\eta_0 \equiv \eta_{\text{cl}} - \omega\beta. \quad (2.26)$$

The rotated fields obey

$$\phi_{\text{rot}}(\tau + \beta, r) = e^{\eta_0} \phi_{\text{rot}}(\tau, r), \quad \bar{\phi}_{\text{rot}}(\tau + \beta, r) = e^{-\eta_0} \bar{\phi}_{\text{rot}}(\tau, r). \quad (2.27)$$

Only the half-interval $\tau \in [-\beta/2, 0]$ is stored. The branch at $\tau > 0$ is reconstructed from the turning-slice symmetry

$$\phi(\tau, r) = \bar{\phi}(-\tau, r), \quad \bar{\phi}(\tau, r) = \phi(-\tau, r), \quad \tau > 0. \quad (2.28)$$

In the rotated variables, the same relation holds for the fluctuations at the turning slice, because the homogeneous rotation has been factored out consistently.

At $\tau = 0$, the ghost point just beyond the stored half-box is obtained by reflection and exchange:

$$y_{j,-1} = \bar{y}_{j,0}, \quad \bar{y}_{j,-1} = y_{j,0}. \quad (2.29)$$

This is the discrete version of the continuum turning conditions

$$\phi(0, r) = \bar{\phi}(0, r), \quad \partial_\tau \phi(0, r) = -\partial_\tau \bar{\phi}(0, r). \quad (2.30)$$

At the past Euclidean boundary, the ghost relation combines the full-box twist with the reflection-and-exchange symmetry. Consider a point just outside the stored domain,

$$\tau_g = -\frac{\beta}{2} - \delta, \quad \delta = \frac{\Delta\tau}{2}. \quad (2.31)$$

Then $\tau_g + \beta = \beta/2 - \delta$, which lies in the reconstructed half. Using first the twist and then the reflection relation gives the half-box ghost rule. In terms of the total rotated fields, this is

$$y_{\text{tot},j,N_\tau} = e^{-\eta_0} \bar{y}_{\text{tot},j,N_\tau-1}, \quad \bar{y}_{\text{tot},j,N_\tau} = e^{+\eta_0} y_{\text{tot},j,N_\tau-1}. \quad (2.32)$$

Equivalently, in terms of the stored fluctuations,

$$y_{j,N_\tau} = e^{-\eta_0} (\bar{y}_{j,N_\tau-1} + r_j \varphi_i) - r_j \varphi_i, \quad (2.33)$$

$$\bar{y}_{j,N_\tau} = e^{+\eta_0} (y_{j,N_\tau-1} + r_j \varphi_i) - r_j \varphi_i. \quad (2.34)$$

These affine ghost relations are the discrete implementation of the residual fixed-charge twist on the half-box.

The regularized variables already encode regularity at the origin. At the outer radial boundary, the production solver imposes Neumann conditions on the rotated fields,

$$\partial_r \phi_{\text{rot}}(L_r, \tau) = 0, \quad \partial_r \bar{\phi}_{\text{rot}}(L_r, \tau) = 0. \quad (2.35)$$

Using (2.14), these become

$$\partial_r y(L_r, \tau) - \frac{1}{L_r} y(L_r, \tau) = 0, \quad \partial_r \bar{y}(L_r, \tau) - \frac{1}{L_r} \bar{y}(L_r, \tau) = 0. \quad (2.36)$$

The corresponding one-sided radial ghost prescription is

$$y_{N_r,i}^{\text{ghost}} = \left(1 + \frac{\Delta r}{L_r}\right) y_{N_r-1,i}, \quad \bar{y}_{N_r,i}^{\text{ghost}} = \left(1 + \frac{\Delta r}{L_r}\right) \bar{y}_{N_r-1,i}. \quad (2.37)$$

2.4 Newton solve and fixed-charge outer loop

All interior values of y and $\bar{y}$ are packed into the real vector $\mathbf{x}$ in (2.18). The nonlinear residual $\mathbf{F}(\mathbf{x})$ is the real and imaginary split of Eqs. (2.22)–(2.23), evaluated with the ghost rules described above. Newton's method solves

$$\mathbf{J}(\mathbf{x}^{(k)}) \delta \mathbf{x}^{(k)} = -\mathbf{F}(\mathbf{x}^{(k)}), \quad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha_k \delta \mathbf{x}^{(k)}, \quad (2.38)$$

where

$$\mathbf{J} = \frac{\partial \mathbf{F}}{\partial \mathbf{x}} \quad (2.39)$$

is an analytic sparse Jacobian, and $\alpha_k \in (0, 1]$ is chosen by a backtracking line search. Because the residual equations contain only nearest-neighbour finite differences, the Jacobian has the usual

local stencil structure. In the real split, each grid point carries a 4×4 local block corresponding to

$$(\operatorname{Re} y, \operatorname{Im} y, \operatorname{Re} \bar{y}, \operatorname{Im} \bar{y}), \quad (2.40)$$

with additional on-site couplings from the nonlinear dependence of $A(u)$ on $y_{\text{tot}} \bar{y}_{\text{tot}}$.

The fixed-charge problem is solved by nesting this Newton solve inside an outer one-dimensional root search for η_0 . For a trial value of η_0 , the inner Newton solve returns a candidate saddle satisfying the bulk equations and all boundary conditions. The charge of that configuration is then evaluated at the turning slice.

The Euclidean charge density used in the code is

$$q_E(\tau, r) = \bar{\phi} \partial_\tau \phi - \phi \partial_\tau \bar{\phi}, \quad (2.41)$$

and the total charge is

$$Q(\tau) = 4\pi \int_0^{L_r} dr r^2 q_E(\tau, r). \quad (2.42)$$

Since only the half-box is stored, the derivatives at the turning slice are reconstructed with the same reflection-and-exchange ghost rules used in the Euclidean residual. This gives a charge $Q_{\text{bounce}}(\eta_0)$ for the candidate bounce.

For the homogeneous branch $y = \bar{y} = 0$, one obtains

$$Q_i = 4\pi \int_0^{L_r} dr r^2 2\omega \varphi_i^2 = \frac{8\pi}{3} \omega \varphi_i^2 L_r^3 = \frac{4\pi}{3} \omega \rho_i^2 L_r^3. \quad (2.43)$$

The fixed-charge condition is imposed by solving

$$\mathcal{Q}(\eta_0) \equiv Q_{\text{bounce}}(\eta_0) - Q_{\text{target}} = 0. \quad (2.44)$$

In practice, the root search in η_0 is performed by continuation in Q_{target} , with each Newton solve warm-started from the previously converged configuration. This continuation is important for tracking the nontrivial branch and avoiding convergence back to the homogeneous solution.

3 Ansatz construction

The two-dimensional fixed- Q equations define a nonlinear saddle-point problem. Newton iteration is local, so convergence to the desired branch requires a seed that is already sufficiently close to it. We therefore construct deterministic initial profiles from one-dimensional solutions and embed them into the two-dimensional (r, τ) domain.

All seed profiles are ultimately passed to the solver in terms of the variables $y, \bar{y}$ defined in Eq. (2.13). If a one-dimensional profile is obtained in terms of the physical modulus $\rho_{1d}(s)$, we first convert it to solver units,

$$\varphi_{1d}(s) = \frac{\rho_{1d}(s)}{\sqrt{2}}. \quad (3.1)$$

The asymptotic value of each profile is the homogeneous background

$$\varphi_i = \lim_{s \rightarrow \infty} \varphi_{1d}(s). \quad (3.2)$$

Once a two-dimensional seed $\phi_{2d}(r, \tau)$ has been built, we initialise

$$y(r, \tau) = r [\phi_{2d}(r, \tau) - \varphi_i], \quad \bar{y}(r, \tau) = y(r, \tau). \quad (3.3)$$

The equality $y = \bar{y}$ is only an initial choice for the seed; the Newton solve then treats the two fields independently.

3.1 $O(4)$ seed

The bounce-like branch is seeded from the one-dimensional $d = 4$ profile. We embed it through the Euclidean radius

$$s(r, \tau) = \sqrt{r^2 + \left(\frac{\tau}{\alpha_\tau}\right)^2}, \quad (3.4)$$

where α_τ allows a mild anisotropic rescaling between the radial and Euclidean-time directions. This gives a localized configuration centred near the turning slice.

To avoid introducing a large deformation near the past Euclidean boundary, only the deviation from the homogeneous tail is windowed. With

$$T = \frac{\beta}{2}, \quad \Delta = f_{\text{win}} T, \quad (3.5)$$

we use

$$w_\tau^{\text{O4}}(\tau) = \begin{cases} 1, & -\Delta \leq \tau \leq 0, \\ \frac{1}{2} \left[1 + \cos\left(\pi \frac{\tau + \Delta}{T - \Delta}\right) \right], & -T \leq \tau < -\Delta. \end{cases} \quad (3.6)$$

The resulting two-dimensional seed is

$$\phi_{2d}^{(4)}(r, \tau) = \varphi_i + w_\tau^{\text{O4}}(\tau) \left[\varphi_{1d}^{(4)}(s(r, \tau)) - \varphi_i \right], \quad (3.7)$$

and the solver variables are obtained from (3.3).

3.2 $O(3)$ static seed

The static branch is seeded from the one-dimensional $d = 3$ critical-bubble profile. This profile is copied unchanged on every Euclidean-time slice:

$$\phi_{2d}^{(3)}(r, \tau) = \varphi_i + A_{\text{seed}} \left[\varphi_{1d}^{(3)}(r) - \varphi_i \right], \quad (3.8)$$

where A_{seed} is an optional amplitude parameter. This seed has no explicit τ -dependence and is used to probe the $O(3)$ -like static branch of the equations.

3.3 $O(1)$ Euclidean-time seed

The $O(1)$ -type seed is constructed from a one-dimensional profile along the Euclidean-time direction. We evaluate

$$s = |\tau| \quad (3.9)$$

and copy the resulting deformation uniformly in the radial direction:

$$\phi_{2d}^{(1)}(r, \tau) = \varphi_i + A_{\text{seed}} \left[\varphi_{1d}^{(1)}(|\tau|) - \varphi_i \right]. \quad (3.10)$$

This seed is useful for identifying branches whose dominant structure is localized in Euclidean time rather than in the radial direction.

3.4 Homogeneous seed and solver input

For comparison, we also use the homogeneous seed

$$\phi_{2d}^i(r, \tau) = \varphi_i, \quad (3.11)$$

which corresponds to

$$y^i(r, \tau) = 0, \quad \bar{y}^i(r, \tau) = 0. \quad (3.12)$$

After any of the seeds above is constructed, it is converted to $y, \bar{y}$, packed into the real vector $\mathbf{x}$, and used as the initial iterate for the Newton solver. No additional explicit deformation along a negative mode is added in the implementation described here.

4 Post-tunnelling Minkowski evolution

We finally describe how the Euclidean fixed- Q saddle is continued to real time. The aim is twofold. First, the turning slice of the Euclidean solution provides initial data for the post-nucleation bubble. Second, the subsequent real-time evolution gives a direct check that the Euclidean saddle lies on a physically meaningful decay branch.

The evolution is performed directly in terms of the complex fields $\phi(t, r)$ and $\bar{\phi}(t, r)$, rather than in polar variables. This avoids coordinate singularities when the modulus becomes small. The physical modulus and phase can then be reconstructed for diagnostics whenever the solution remains close to the physical subspace $\bar{\phi} = \phi^*$.

4.1 Initial data from the turning slice

The Euclidean solution is computed on

$$\tau \in \left[-\frac{\beta}{2}, 0\right], \quad r \in [0, L_r], \quad (4.1)$$

with the turning slice at $\tau = 0$. At the continuum level, the turning conditions are

$$\phi(0, r) = \bar{\phi}(0, r), \quad \partial_\tau \phi(0, r) = -\partial_\tau \bar{\phi}(0, r). \quad (4.2)$$

In the code, the derivatives at $\tau = 0$ are reconstructed using the same ghost rules employed by the Euclidean solver, so that the real-time initial data are exactly consistent with the discretized Euclidean saddle.

Let i_0 denote the stored Euclidean-time slice closest to the turning surface. From the converged solution, one obtains

$$\phi_0(r) \equiv \phi(\tau_{i_0}, r), \quad \bar{\phi}_0(r) \equiv \bar{\phi}(\tau_{i_0}, r), \quad (4.3)$$

and the corresponding Euclidean derivatives

$$\phi_{\tau,0}(r) \equiv \partial_\tau \phi|_{\tau=\tau_{i_0}}, \quad \bar{\phi}_{\tau,0}(r) \equiv \partial_\tau \bar{\phi}|_{\tau=\tau_{i_0}}. \quad (4.4)$$

With our Wick-rotation convention, the Minkowski initial velocities are

$$\dot{\phi}_0(r) = i \phi_{\tau,0}(r), \quad \dot{\bar{\phi}}_0(r) = i \bar{\phi}_{\tau,0}(r). \quad (4.5)$$

The Cauchy data for the real-time evolution are therefore

$$\phi(0, r) = \phi_0(r), \quad \bar{\phi}(0, r) = \bar{\phi}_0(r), \quad \dot{\phi}(0, r) = \dot{\phi}_0(r), \quad \dot{\bar{\phi}}(0, r) = \dot{\bar{\phi}}_0(r). \quad (4.6)$$

4.2 Real-time equations

The Minkowski equations are evolved directly for ϕ and $\bar{\phi}$:

$$\ddot{\phi} = \nabla_r^2 \phi - \frac{\partial U}{\partial \phi}, \quad \ddot{\bar{\phi}} = \nabla_r^2 \bar{\phi} - \frac{\partial U}{\partial \bar{\phi}}, \quad (4.7)$$

where

$$\nabla_r^2 f = \partial_r^2 f + \frac{2}{r} \partial_r f. \quad (4.8)$$

Since $U = U(\rho)$ and $\rho = \sqrt{2\phi\bar{\phi}}$, one has

$$\frac{\partial U}{\partial \bar{\phi}} = \frac{dU}{d\rho} \frac{\phi}{\rho}, \quad \frac{\partial U}{\partial \phi} = \frac{dU}{d\rho} \frac{\bar{\phi}}{\rho}. \quad (4.9)$$

Introducing

$$C(\phi, \bar{\phi}) \equiv \frac{1}{\rho} \frac{dU}{d\rho}, \quad (4.10)$$

the equations take the compact form

$$\ddot{\phi} = \nabla_r^2 \phi - C \phi, \quad \ddot{\bar{\phi}} = \nabla_r^2 \bar{\phi} - C \bar{\phi}. \quad (4.11)$$

4.3 Radial discretization

The real-time radial grid is

$$0 = r_0 < r_1 < \cdots < r_{N_r-1} = L_r. \quad (4.12)$$

The spherical Laplacian is discretized in finite-volume form, using shell boundaries

$$r_{i+\frac{1}{2}} = \frac{1}{2}(r_i + r_{i+1}) \quad (4.13)$$

and shell volumes

$$\mathcal{V}_i = \frac{4\pi}{3} \left(r_{i+\frac{1}{2}}^3 - r_{i-\frac{1}{2}}^3 \right). \quad (4.14)$$

For a field $f(r)$, define the edge gradients

$$g_{i+\frac{1}{2}} = \frac{f_{i+1} - f_i}{r_{i+1} - r_i}, \quad (4.15)$$

and the corresponding fluxes

$$\mathcal{F}_{i+\frac{1}{2}} = 4\pi r_{i+\frac{1}{2}}^2 g_{i+\frac{1}{2}}. \quad (4.16)$$

The discrete Laplacian is then

$$\mathcal{V}_i (\nabla_r^2 f)_i = \mathcal{F}_{i+\frac{1}{2}} - \mathcal{F}_{i-\frac{1}{2}}. \quad (4.17)$$

At the origin, the left flux vanishes because the area factor goes to zero, which enforces regularity. This finite-volume form is chosen so that the same edge gradients entering the spatial operator also define the gradient contribution to the discrete Hamiltonian.

4.4 Time integrator and boundary conditions

The time evolution uses a second-order explicit Störmer–Verlet, or velocity-Verlet, integrator [2, 3]. If a_ϕ^n and $a_{\bar{\phi}}^n$ are the accelerations computed from (4.11) at time t_n , the update is

$$\phi^{n+1} = \phi^n + \Delta t \dot{\phi}^n + \frac{1}{2} \Delta t^2 a_\phi^n, \quad (4.18)$$

$$\bar{\phi}^{n+1} = \bar{\phi}^n + \Delta t \dot{\bar{\phi}}^n + \frac{1}{2} \Delta t^2 a_{\bar{\phi}}^n, \quad (4.19)$$

$$\dot{\phi}^{n+\frac{1}{2}} = \dot{\phi}^n + \frac{1}{2} \Delta t a_\phi^n, \quad (4.20)$$

$$\dot{\bar{\phi}}^{n+\frac{1}{2}} = \dot{\bar{\phi}}^n + \frac{1}{2} \Delta t a_{\bar{\phi}}^n. \quad (4.21)$$

After recomputing the accelerations at the new fields, the velocities are completed as

$$\dot{\phi}^{n+1} = \dot{\phi}^{n+\frac{1}{2}} + \frac{1}{2} \Delta t a_\phi^{n+1}, \quad (4.22)$$

$$\dot{\bar{\phi}}^{n+1} = \dot{\bar{\phi}}^{n+\frac{1}{2}} + \frac{1}{2} \Delta t a_{\bar{\phi}}^{n+1}. \quad (4.23)$$

In the runs described in the main text we use $m\Delta t = 10^{-4}$, evolve up to a prescribed $t_{\max}$, and store snapshots at selected times.

At the origin, regularity is imposed by copying the first interior point,

$$\phi_0 = \phi_1, \quad \bar{\phi}_0 = \bar{\phi}_1, \quad \dot{\phi}_0 = \dot{\phi}_1, \quad \dot{\bar{\phi}}_0 = \dot{\bar{\phi}}_1. \quad (4.24)$$

At the outer boundary, no additional projection, pinning, or absorbing condition is applied in the production runs discussed here; the boundary values are evolved by the same update as the bulk points. We also do not explicitly project the solution back onto $\bar{\phi} = \phi^*$ after each timestep. The data extracted from the turning slice starts on the physical branch, and the evolution remains close to it within numerical accuracy.

4.5 Energy and charge diagnostics

The main diagnostics are the total charge and energy. The Minkowski charge density is

$$q_M(r, t) = -i \left(\bar{\phi} \dot{\phi} - \dot{\bar{\phi}} \phi \right), \quad (4.25)$$

and

$$Q(t) = 4\pi \int_0^{L_r} dr r^2 q_M(r, t). \quad (4.26)$$

On the physical subspace,

$$\phi = \frac{\rho}{\sqrt{2}} e^{i\theta}, \quad \bar{\phi} = \phi^*, \quad (4.27)$$

this reduces to

$$q_M = \rho^2 \dot{\theta}. \quad (4.28)$$

The energy is computed in a form consistent with the finite-volume discretization. In continuum notation, the Hamiltonian is

$$E_M = 4\pi \int_0^{L_r} dr r^2 \left[\dot{\phi} \dot{\bar{\phi}} + \partial_r \phi \partial_r \bar{\phi} + U(\rho) \right]. \quad (4.29)$$

For a physical configuration this is equivalent to

$$E_M = 4\pi \int_0^{L_r} dr r^2 \left[\frac{1}{2} \dot{\rho}^2 + \frac{1}{2} (\partial_r \rho)^2 + \frac{1}{2} \rho^2 \dot{\theta}^2 + \frac{1}{2} \rho^2 (\partial_r \theta)^2 + U(\rho) \right]. \quad (4.30)$$

Monitoring (4.26) and (4.29) provides the primary check that the real-time continuation of the Euclidean saddle is numerically stable and physically consistent.

References

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